

NYU SPS Agentic AI Recruiting Assistant:

Building an AI-Powered System to Support Prospective Student Inquiries and Reduce Faculty Administrative Burden

Applied Project Final Report

By
Yash Sharma
Spring, 2026

A paper submitted in partial fulfillment of the requirements for the degree of

Master of Science in Management and Analytics

at the

Division of Programs in Business
School of Professional Studies
New York University

Project Sponsor: Dr. Andres Fortino, Clinical Associate Professor, NYU SPS

Presentation: <https://docs.google.com/presentation/d/1mi8bl24iT1cyxjs0NModh3HzG1wfWw6cLJVHeOWxis/edit?usp=sharing>

GitHub Repository: <https://github.com/yash-sharma/nyu-sps-bobcat>

Table of Contents

Declaration	4
Acknowledgments	5
Abstract	6
Abbreviations and Definitions	7
1. Introduction	8
2. Project Objectives and Metrics	11
3. Alternate Solutions Evaluated	14
4. Literature Survey	17
5. Approach and Methodology	24
6. Results	31
7. Issues Encountered	37
8. Lessons Learned	40
9. Conclusion and Further Work	42
References	46
Appendix A – Project Acceptance Document	49
Appendix B – Project Sponsor Agreement	50
Appendix C – Functional Requirements Specifications	51
Appendix D – Project Plan (WBS)	52
Appendix E – Risk Management Plan	53
Appendix F – Technology Trial Plan	54
Appendix G – Status Reports	55
Appendix H – Annotated Bibliography	56

Declaration

I, Yash Sharma, declare that this project report submitted by me to the School of Professional Studies, New York University, in partial fulfillment of the requirement for the award of the degree of Master of Science in Management and Systems, is a record of project work carried out by me under the guidance of Dr. Andres Fortino, NYU Clinical Associate Professor of Management and Systems.

I grant powers of discretion to the Division of Programs in Business, School of Professional Studies, and New York University to allow this report to be copied in part or in full without further reference to me. The permission covers only copies made for study purposes or for inclusion in Division of Programs in Business, School of Professional Studies, and New York University research publications, subject to normal conditions of acknowledgment.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Signature:  _____

Name: Yash Sharma

Date: April 29, 2026

Acknowledgments

I sincerely thank Dr. Andres Fortino, Clinical Associate Professor and Faculty Advisor at the NYU School of Professional Studies, for his invaluable guidance, mentorship, and sponsorship throughout this project. His vision for applied AI in higher education and his commitment to rigorous academic standards shaped every phase of this work. I also extend my gratitude to the NYU SPS Office of Admissions for providing the institutional context, program information, and stakeholder access that made the proof-of-concept prototype both feasible and relevant.

I want to thank all the instructors in the Management and Systems program from whom I have taken courses and learned a great deal. The interdisciplinary foundation in systems thinking, business analytics, technology management, and organizational behavior that this program provided proved essential to delivering a project that bridges technical and human dimensions of a real institutional challenge.

I am also grateful to Digital Forge, the technology innovation partner that co-sponsored this engagement, for providing the professional framework and industry perspective that elevated this capstone beyond a purely academic exercise. The Fall 2025 Advanced Business Analytics Team, whose foundational needs assessment survey established the quantitative baseline for this project, deserves particular acknowledgment.

Finally, I acknowledge the use of Claude (Anthropic) as an AI-assisted drafting tool for portions of the technology trial plan and literature review synthesis, in accordance with the program's documentation requirements for LLM use disclosure.

Abstract

Graduate program recruiting at NYU School of Professional Studies (SPS) faces a critical operational challenge: faculty and staff devote substantial time answering repetitive prospective student inquiries, with 40% of faculty engaging with prospective students multiple times per week and 80% reporting inconsistent information delivery across the institution. This project proposes and validates an agentic AI recruiting assistant — named Bobcat — as a proof-of-concept solution to automate, standardize, and improve the prospective student inquiry experience at NYU SPS.

The project employed a design-build-validate methodology structured across four SMART objectives: requirements definition and knowledge base construction, prototype development, usability validation, and platform finalization with documentation for institutional handoff. The core technology is an agentic AI system built on the Anthropic Claude API (claude-sonnet-4), implemented as a React-based web interface with a FastAPI backend, Pinecone vector database for retrieval-augmented generation (RAG), and seven specialized tool-calling agents covering program matching, prerequisite assessment, deadline tracking, admissions guidance, resume analysis, career services information, and human escalation routing.

The prototype achieved 88% response accuracy on a faculty-validated 50-question test set, surpassing the 85% project threshold. A usability study with 7 faculty and admissions staff participants confirmed the system's practical utility. The A/B technology trial demonstrated a first-response time reduction from 24 hours (manual) to under 10 seconds (AI), with an escalation rate of 9%, within the target range. The proof-of-concept was completed on schedule within a 300-hour student effort budget and all four project objectives were met or exceeded.

This project demonstrates that agentic AI systems can meaningfully address administrative burden in graduate program recruiting through automated, accurate, and compliant conversational interactions. The architecture is directly replicable across other NYU schools and provides a scalable framework for higher education institutions facing similar operational challenges. Future work should focus on production-grade FERPA compliance certification, CRM integration, bias testing across diverse student profiles, and a phased student-facing pilot deployment.

Keywords: agentic AI, large language models, retrieval-augmented generation, higher education, recruiting assistant, chatbot, FERPA compliance, faculty administrative burden, proof of concept, NYU SPS

Abbreviations and Definitions

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
ECTM	Experimentum Crucis Technology Matrix
FERPA	Family Educational Rights and Privacy Act
FRS	Functional Requirements Specification
LLM	Large Language Model
NYU	New York University
NYU SPS	NYU School of Professional Studies
PICO	Population, Intervention, Comparison, Outcome (trial framework)
RAG	Retrieval-Augmented Generation
SMART	Specific, Measurable, Achievable, Relevant, Time-bound
SPS	School of Professional Studies
TTF	Task-Technology Fit
UI	User Interface
WCAG	Web Content Accessibility Guidelines
WBS	Work Breakdown Structure

1. Introduction

The application of artificial intelligence to institutional administrative processes represents one of the most consequential and practically accessible innovations available to higher education organizations today. Large language models (LLMs), which can engage in natural language conversation, retrieve structured knowledge, and route complex requests to human experts, offer a compelling technology pathway for institutions facing the dual pressure of rising inquiry volumes and constrained administrative capacity. This project explores that pathway through the design, development, and validation of an agentic AI recruiting assistant for the NYU School of Professional Studies.

Problem

Graduate program recruiting at NYU SPS involves a significant and growing administrative burden. A Fall 2025 faculty needs assessment survey conducted by the Advanced Business Analytics Team revealed that 40% of SPS faculty communicate with prospective students multiple times per week. More troublingly, 80% of faculty reported that inconsistent information is shared across different faculty members, creating confusion for prospective students making high-stakes educational decisions. Approximately 60% of all prospective student inquiries fall into predictable, automatable categories: admissions procedures, course difficulty assessments, prerequisite requirements, and program fit guidance.

This pattern generates four compounding institutional problems. First, highly trained faculty and professional staff allocate substantial time to tasks that do not require their expertise. Second, inconsistent messaging across faculty creates a fractured information environment that disadvantages applicants. Third, the absence of 24/7 availability places students in different time zones or with demanding work schedules at a systematic disadvantage. Fourth, the lack of structured data collection about inquiry patterns means that common points of student confusion remain invisible to program administrators.

Approach

The proposed solution is a working prototype of an agentic AI recruiting assistant — named Bobcat — that provides automated, accurate, and FERPA-compliant responses to prospective student inquiries using a faculty-approved knowledge base. Bobcat is designed as a first-response layer that handles the automatable 60% of inquiries independently while routing complex, sensitive, or out-of-scope questions to human advisors with full conversational context. The system is not designed to replace human advisors but to optimize their time by eliminating the routine and repetitive first-contact load.

Core Technology

Bobcat is built on the Anthropic Claude API, specifically using the claude-sonnet-4 model, which provides state-of-the-art natural language understanding, tool use, and multi-turn conversation capabilities. The system implements an agentic architecture in which Claude orchestrates seven specialized tools: a Program Matcher, Academic Audit (prerequisite checker), Deadline Tracker, Admissions Guide, Resume Analyzer, Career Services information agent, and an Escalation Handler. This tool-calling architecture allows the system to provide structured, verifiable responses grounded in faculty-approved data rather than raw generative output, significantly reducing hallucination risk and improving factual accuracy.

The front-end is built in React, providing a responsive, NYU SPS-branded chat interface with an analytics dashboard. The back-end uses FastAPI for the API layer and Pinecone for the vector database that powers retrieval-augmented generation (RAG) across the 54-question knowledge base. The system is designed to be deployed as a lightweight JavaScript widget embeddable on any NYU SPS program web page.

Benefits

The projected benefits of Bobcat are measurable and institutional in scope. For faculty and staff, the system reduces the administrative burden of repetitive inquiries, freeing professional time for teaching, research, and complex student advising. For prospective students, it provides instant, 24/7 access to accurate information, improving the inquiry experience regardless of time zone or schedule constraints. For NYU SPS as an institution, it delivers consistent messaging across all prospective student interactions and generates structured analytics about inquiry patterns that can inform program communications and marketing strategy. The system also positions NYU SPS as an innovation leader in graduate admissions technology.

Research Question

This project addresses the following research question: Can an agentic AI system, built on a faculty-approved knowledge base and Claude LLM, achieve the accuracy, usability, and compliance standards necessary to serve as a reliable first-response layer for prospective student inquiries at NYU School of Professional Studies?

Contribution

This project makes three primary contributions. It delivers a functional, validated proof-of-concept prototype demonstrating that agentic AI can achieve the performance thresholds (>85% accuracy, <15% escalation rate, >3.5/5.0 user satisfaction) required for institutional deployment. It produces a replicable technical architecture and

documentation package — including a FERPA compliance framework, technical architecture guide, and operations manual — suitable for adaptation by other NYU schools or higher education institutions facing equivalent challenges. Finally, it contributes a structured evaluation methodology (TTF analysis, ECTM scoring, A/B trial design) for assessing the readiness of AI technologies for higher education administrative applications.

Sponsor

This project was sponsored by Dr. Andres Fortino, Clinical Associate Professor and Faculty Advisor at the NYU School of Professional Studies, 7 East 12th Street, New York, NY 10003. Digital Forge, a technology innovation partner collaborating with NYU SPS on AI-driven educational solutions, served as the institutional co-sponsor. The project was conducted as a capstone engagement within the Master of Science in Management and Systems program (MASY GC-4100) for the Spring 2026 semester.

Importance of Project

For Digital Forge and NYU SPS, the completion of this project represents a direct response to a documented and quantified operational pain point. The needs assessment baseline — 40% weekly faculty contact, 80% information inconsistency, 60% automatable inquiry rate — establishes a clear and measurable problem. Bobcat addresses this problem with a technology that is commercially proven at scale and specifically adapted to the institutional context of NYU SPS. The sponsor's primary interest is in validating that an AI-powered approach can achieve the accuracy and usability thresholds necessary to justify a broader pilot deployment across the institution's 20+ graduate programs. Successfully completing this proof of concept creates a direct path to operational deployment and positions NYU SPS as a first-mover in AI-assisted graduate admissions.

2. Project Objectives and Metrics

Goal of the Project

The goal of the project was to design, develop, and validate a working prototype of an agentic AI recruiting assistant for NYU SPS that demonstrates technical feasibility, faculty acceptance, and measurable value in reducing administrative burden while improving the prospective student experience.

Project Deliverables and Metrics

Objective 1 – Define project scope, requirements, and build a faculty-approved knowledge base.

Deliverable: A requirements specification document and a knowledge base covering 50 or more common questions across key recruiting categories (admissions procedures, program descriptions, prerequisites, and application processes).

Metric: Sponsor approval and faculty review sign-off on the requirements specification and knowledge base by Week 4 (end of February 2026).

Status: COMPLETED — EXCEEDED

The knowledge base was completed with 54 faculty-approved Q&A pairs (exceeding the 50-question target) by Week 4. The requirements specification document was approved by the project sponsor.

Objective 2 – Develop a working prototype with conversational AI and web interface.

Deliverable: A functional web-based chat interface integrated with an LLM backend, escalation logic for routing complex queries to human advisors, and NYU SPS branding.

Metric: Functional demonstration achieving greater than 85% accuracy on a test set of 50 representative prospective student questions by Week 8 (end of March 2026).

Status: COMPLETED — EXCEEDED

The prototype achieved 88% response accuracy on the 50-question test set (exceeding the 85% target) by Week 8. The system includes a React front-end, FastAPI backend, Pinecone vector database, seven specialized tool agents, WCAG 2.1 AA compliance, and full NYU SPS branding. A/B technology trial was completed per the approved trial plan.

Objective 3 – Validate prototype through testing and usability studies.

Deliverable: A validation report containing testing results, usability findings from 5 to 10 prospective faculty or admissions staff participants, and documented escalation testing outcomes.

Metric: Faculty acceptance and sponsor approval of the validation report by Week 10 (mid-April 2026).

Status: COMPLETED

The validation study was completed with 7 faculty and admissions staff participants (within the 5–10 target range) by Week 10. Usability findings and escalation testing outcomes were documented and the validation report received sponsor approval.

Objective 4 – Finalize platform, analytics dashboard, and documentation for handoff.

Deliverable: A complete system with an analytics dashboard, FERPA compliance documentation, technical architecture guide, operations manual, and a final capstone report of 10 to 12 pages.

Metric: Sponsor sign-off, GitHub repository validation by faculty and admissions, and project completion by Week 12 (end of Spring 2026 semester, May 2026).

Status: COMPLETED

All documentation was finalized by Week 12 and the GitHub repository was validated. The analytics dashboard includes eight KPI metric cards, monthly inquiry volume charts, user satisfaction visualizations, program interest rankings, lead conversion funnel analysis, and a FERPA compliance audit panel. The only outstanding item at project close is sponsor signature on the final acceptance document, pending resolution of the faculty strike.

Project Evaluation

Project success was evaluated against the four SMART objectives described above, using the metrics established in the approved project proposal (Assignment 1B) and ratified by the project sponsor in the signed Sponsor Acceptance Document (Appendix A). The primary quantitative evaluation criteria were: knowledge base coverage (54/50 questions, 108% of target), prototype accuracy (88%, exceeding the 85% threshold), usability study participant count (7, within the 5–10 target), and on-time delivery of all four milestones within the 300-hour budget.

3. Alternate Solutions Evaluated

Before committing to the agentic AI architecture, three alternative solution approaches were systematically evaluated against defined criteria. Each alternative represented a real option available to NYU SPS for addressing the prospective student inquiry burden problem.

Solution Evaluation Criteria

The following criteria were established to evaluate and compare potential solutions:

- **Response Accuracy:** The ability of the solution to provide correct, faculty-approved information to prospective students across the full range of common inquiry categories.
- **Scalability:** The solution's ability to handle increasing inquiry volume without proportional increases in cost or staff time.
- **Availability:** Whether the solution provides 24/7 access regardless of staff working hours or time zones.
- **Consistency:** The degree to which the solution delivers uniform, standardized information across all interactions.
- **Implementation Cost and Complexity:** The technical and organizational investment required to implement and maintain the solution.
- **FERPA Compliance:** Whether the solution can be deployed in a manner consistent with student data privacy requirements.
- **Faculty Acceptance:** The likelihood that faculty and admissions staff will accept and utilize the solution.

Alternative Solution A: Enhanced FAQ / Static Knowledge Base

The simplest alternative was to invest in a comprehensive static FAQ page or PDF document covering common prospective student questions. This approach requires no custom technology development, minimal cost, and straightforward content governance. Faculty would approve content during an initial development phase, and updates would be made periodically by admissions staff.

However, this solution fails on several evaluation criteria. Static FAQs do not support conversational interaction, meaning students must browse or search for answers rather than asking questions naturally. They provide no escalation pathway, no personalization, no analytics about inquiry patterns, and no 24/7 interactive availability beyond passive page viewing. Most critically, static FAQs do not address the information inconsistency problem — faculty would still receive questions directly from students who prefer human contact or cannot find their answer in the FAQ. Research consistently shows that static FAQ pages achieve adoption rates of 20–30% for self-service resolution, leaving 70–80%

of inquiries to reach staff. This alternative was rejected for failing to achieve scalability, consistency, or meaningful burden reduction.

Alternative Solution B: Rule-Based Chatbot (Scripted Decision Tree)

A rule-based or scripted chatbot represents the most common entry point for institutional chatbot deployment. These systems use predefined decision trees, keyword matching, and scripted responses to guide users through inquiry flows. Several commercial platforms (Intercom, Drift, Tidio) offer no-code or low-code rule-based chatbot builders specifically designed for higher education admissions.

This approach offers faster implementation than a custom AI system, lower initial cost, and easier content governance through visual workflow editors. It does not require LLM API access, reducing ongoing operational costs and eliminating concerns about generative AI hallucination.

However, rule-based chatbots are fundamentally limited by their inability to handle the natural language variability of prospective student questions. Graduate students rarely ask questions using precisely the keywords that map to a decision tree branch. When questions fall outside the scripted flows — which research suggests occurs in 30–40% of real-world interactions — rule-based systems produce unhelpful dead-ends or redirect users to human staff, negating the scalability benefit. They also cannot perform the kind of multi-step reasoning required for tasks like prerequisite assessment (evaluating a student's background against specific program requirements) or program matching (scoring multiple programs against a student's stated goals). This alternative was rejected for failing to achieve the accuracy and conversational flexibility thresholds required.

Alternative Solution C: Expanded Human Staff Capacity

The most direct alternative to AI automation is simply hiring additional admissions staff or reallocating faculty time to formalize and expand human response capacity. This could take the form of a dedicated admissions inquiry team, extended office hours, or a formalized peer-advisor program using current students.

This approach offers the highest possible response quality and complete conversational flexibility, with no technical risk and full FERPA compliance through existing institutional frameworks. For certain high-stakes inquiry categories — visa questions, financial aid appeals, disability accommodations — human judgment is essential and cannot be substituted by any AI system.

However, this alternative does not address the structural problem. Adding human staff is expensive (a full-time admissions coordinator costs approximately \$55,000–\$75,000 annually in the New York market), does not provide 24/7 availability, does not solve the information inconsistency problem (different staff members still give different answers without a centralized knowledge management system), and does not scale efficiently with enrollment growth. The needs assessment data — 40% weekly faculty contact, 60% automatable inquiry rate — indicates that the problem is structural, not a capacity gap that additional headcount can solve without process redesign. This alternative was rejected for failing scalability and availability criteria and for offering poor return on investment relative to the AI approach.

Selection Rationale

The agentic AI approach was selected for three primary reasons. First, it is the only solution that simultaneously achieves high accuracy, 24/7 availability, consistent messaging, and meaningful scalability. The RAG architecture grounds responses in faculty-approved content, eliminating the hallucination risk of a purely generative LLM while providing the conversational flexibility that rule-based systems cannot match.

Second, the Task-Technology Fit (TTF) analysis conducted as part of this project (Assignment 3B) confirmed that the specific characteristics of the prospective student inquiry task — high frequency, high criticality, structured data requirements, and moderate complexity — align strongly with the capabilities of agentic AI technology. The TTF score of 3.04/4.00 (Good Fit) and ECTM score of 76.7% (Conditional Approval) provided quantitative validation for proceeding.

Third, the commercial maturity of the underlying technology stack (Claude API, Pinecone, React, FastAPI) ensured that implementation risk was manageable within the capstone timeline. The 88% accuracy result achieved during development validated the selection rationale: the technology performed at the level required for institutional deployment.

4. Literature Survey

Introduction

This literature review examines published research across five thematic areas relevant to the development of an AI-powered recruiting assistant for NYU School of Professional Studies: the higher education admissions industry, the administrative burden problem facing faculty and staff, AI chatbot solutions for student services, the specific technologies enabling retrieval-augmented generation, and documented use cases of similar deployments. The scope is limited to peer-reviewed publications, institutional reports, and foundational technical papers published between 2019 and 2025.

This literature review was created to support the proposed proof of concept and technology trial for the NYU SPS Agentic AI Recruiting Assistant. The review is organized thematically, beginning with the industry context and narrowing toward the specific technologies and use cases that inform the project design. The goal is to demonstrate that the proposed solution is grounded in established research, that the technology has been shown to work in comparable settings, and that the project addresses a well-documented problem with a viable approach.

The Industry

Higher education in the United States is a large and competitive industry, with over 4,000 degree-granting institutions serving approximately 19.4 million students as of the 2023–2024 academic year. Graduate programs represent a particularly competitive segment, as institutions vie for a shrinking pool of prospective students amid demographic shifts and changing workforce expectations. The admissions process is a critical touchpoint where institutions must deliver timely, accurate, and personalized information to prospective students who are often evaluating multiple programs simultaneously. The quality of the inquiry experience directly influences enrollment decisions.

NYU School of Professional Studies occupies a distinctive position within this competitive landscape. The school serves working professionals seeking graduate credentials in management, analytics, technology, real estate, hospitality, global affairs, and related fields across more than 20 graduate programs. Its student population is disproportionately geographically distributed, time-constrained, and accustomed to digital-first information experiences. These characteristics make the inquiry process particularly sensitive to delays, inconsistencies, or limited availability — and particularly well-suited to AI-assisted first-response solutions.

The Problem

The administrative burden of repetitive student inquiries in higher education is extensively documented in the literature. Okonkwo and Ade-Ibijola (2021) identified administrative burden from repetitive student inquiries as one of the most commonly cited motivations for chatbot deployment across a systematic review of 55 higher education chatbot implementations. Wollny et al. (2021) documented in a comprehensive systematic review that institutions deploying chatbots for administrative support reported significant reductions in staff workload, with the effect size correlating strongly with the proportion of inquiry volume addressed by the chatbot.

The NYU SPS Advanced Business Analytics Team's Fall 2025 needs assessment survey reinforced this literature with institution-specific quantitative evidence: 40% of SPS faculty communicate with prospective students multiple times per week, 80% report inconsistent information delivery across faculty, and approximately 60% of inquiries fall into automatable categories. Huang and Zhang (2023) quantified the potential impact of chatbot automation in their systematic review, finding that administrative chatbots in educational settings can reduce staff workload by 30–50% for routine inquiries while maintaining high user satisfaction. This magnitude of potential impact provides strong justification for pursuing an AI-based solution.

The Proposed Solution

The proposed solution is the development and validation of an agentic AI recruiting assistant that automates responses to routine prospective student inquiries using a faculty-approved knowledge base, while escalating complex or sensitive questions to human advisors. Unlike simple FAQ pages or rule-based chatbots, the proposed system uses retrieval-augmented generation to provide contextually relevant, conversational responses grounded in verified institutional content. The approach is informed by the growing body of literature demonstrating that RAG-based chatbots outperform both pure generative models and traditional rule-based systems for domain-specific question answering (Gao et al., 2024; Swacha & Gracel, 2025).

The literature provides strong support for this approach. Perez et al. (2020) documented the evolution of chatbots in education from simple FAQ systems to sophisticated AI agents and found that chatbots integrated into existing institutional platforms are more effective than standalone tools. Islam et al. (2024) reported that an AI chatbot deployed for student services at Universitas Negeri Semarang reduced average response time from 24 hours to under 2 minutes with an 82% resolution rate — a result that directly informed the performance targets for this project. These findings suggest that AI-powered chatbots can meaningfully address the administrative burden described above while improving the prospective student experience.

Critically, the proposed solution incorporates safeguards recommended by the literature. UNESCO (2021) and the OECD (2023) emphasize that AI systems in education must be

transparent about their capabilities, offer human alternatives, and protect user privacy. The U.S. Department of Education (2022) provides specific guidance on FERPA compliance for technology systems interacting with students. These requirements are embedded in the project design through the chatbot's disclosure of its AI nature, the visible human escalation pathway, and the FERPA-compliant data handling architecture.

The Technology

The technical foundation of the proposed solution is retrieval-augmented generation (RAG), a methodology introduced by Lewis et al. (2020) that combines the fluent natural language generation capabilities of LLMs with a structured retrieval system that grounds responses in verified factual content. In a RAG architecture, user queries are converted to vector embeddings, matched against a pre-indexed knowledge base using approximate nearest-neighbor search, and the retrieved documents are injected into the LLM's context window to generate a grounded, accurate response. This approach directly addresses the hallucination problem that makes pure generative LLMs unsuitable for high-stakes institutional applications.

Subsequent advances have significantly improved RAG performance. Karpukhin et al. (2020) demonstrated that dense passage retrieval using BERT-based encoders substantially outperforms traditional sparse retrieval (BM25) for open-domain question answering, providing the technical basis for the Pinecone vector database used in this project. Reimers and Gurevych (2019) introduced Sentence-BERT (SBERT), which produces semantically meaningful sentence embeddings that enable accurate semantic matching even when query phrasing differs from knowledge base content — a critical capability for handling the natural language variability of prospective student questions. Johnson et al. (2021) developed FAISS, a high-performance vector similarity search library that enables real-time retrieval at scale, underpinning the sub-10-second response time target for this project.

Swacha and Gracel (2025) reviewed 89 RAG-based educational chatbot implementations and found that systems using semantic embedding with dense retrieval consistently achieved higher accuracy than hybrid or sparse retrieval approaches, particularly for domain-specific factual questions. This finding directly informed the technical architecture selection for Bobcat.

Use Cases

The deployment of AI chatbots in higher education admissions and student services contexts has expanded rapidly since 2020. Thway et al. (2024) evaluated the impact of a RAG-based chatbot at Nanyang Technological University (NTU) on student learning outcomes and administrative efficiency, finding that the system achieved 84% resolution rate for course-related inquiries and reduced staff response time by 67% during the

implementation period. Importantly, the NTU deployment demonstrated that student satisfaction with AI-assisted responses was comparable to satisfaction with human responses for factual, structured inquiries, while human advisors maintained significantly higher satisfaction for complex or emotionally sensitive interactions — a finding that directly supports the escalation logic embedded in Bobcat's design.

Islam et al. (2024) documented the deployment of an AI-driven chatbot at Universitas Negeri Semarang specifically for admissions-related student services. The system achieved an 82% resolution rate and reduced average response time from 24 hours to under 2 minutes. The study identified knowledge base coverage as the primary determinant of resolution rate, a finding consistent with the decision to develop a faculty-approved 54-question knowledge base as Objective 1 of this project. The study also found that student trust in AI-generated responses was significantly higher when the system clearly disclosed its AI nature and provided easy access to human escalation — design choices embedded in Bobcat's interface from the first prototype.

Conclusion

The evidence from the literature review confirms that RAG-based AI chatbots offer a viable, well-supported approach to addressing the administrative burden of repetitive prospective student inquiries in higher education. The foundational RAG architecture (Lewis et al., 2020) has been refined through advances in dense passage retrieval, sentence embeddings, and vector similarity search into a mature technology stack that consistently achieves high accuracy in domain-specific applications. Systematic reviews confirm that chatbots reduce administrative workload by 30–50% while maintaining user satisfaction. Deployed use cases at NTU and Universitas Negeri Semarang validate that these systems work in practice with resolution rates exceeding 80%. The ethical and regulatory frameworks from UNESCO, OECD, and the U.S. Department of Education provide a clear foundation for responsible deployment.

The most important design implication from the literature is the centrality of knowledge base quality to system performance. Every high-performing deployment reviewed achieved its accuracy results through rigorous, domain-expert-reviewed content development — a finding that directly motivated the faculty sign-off process embedded in Objective 1 of this project. The literature also consistently supports the hybrid human-AI model, where AI handles routine inquiries and humans handle complex or sensitive cases, as the optimal deployment pattern for high-stakes educational contexts.

5. Approach and Methodology

Problem Statement and Research Question

Graduate program recruiting at NYU SPS creates an unsustainable administrative burden: 40% of faculty engage with prospective students multiple times weekly, 80% report information inconsistency, and 60% of inquiries are repetitive and automatable. This project asks: Can an agentic AI system, built on a faculty-approved knowledge base and the Claude LLM, achieve the accuracy, usability, and compliance standards required to serve as a reliable first-response layer for prospective student inquiries at NYU SPS?

Proof of Concept Approach

The project follows a proof-of-concept methodology structured across four sequentially dependent phases aligned with the four project objectives. The approach is grounded in rapid prototype development, empirical validation, and iterative refinement guided by faculty and sponsor feedback at each milestone gate. Refer to Appendix C for the full Functional Requirements Specification.

Phase 1 (Weeks 1–4) focused on requirements definition and knowledge base construction. The requirements specification was developed by synthesizing the faculty needs assessment findings with program-level information from the NYU SPS website, admissions documentation, and direct sponsor consultation. The knowledge base was built using a structured Q&A format covering four primary inquiry categories: admissions procedures (application process, requirements, deadlines, fees), program descriptions (curriculum, learning outcomes, career pathways), prerequisites (program-specific requirements, waiver processes), and application processes (submission procedures, timeline, decision notifications).

Phase 2 (Weeks 5–8) covered system development. The technical architecture selected was a React front-end with a FastAPI backend, Pinecone vector database, and the Anthropic Claude API for LLM inference. The agentic architecture was implemented using Claude's tool-use capability, which allows the model to call structured functions in response to user queries rather than relying solely on generative output. Seven tool agents were implemented: Program Matcher (program recommendation based on career goals and background), Academic Audit (prerequisite eligibility assessment), Deadline Tracker (application deadline retrieval), Admissions Guide (admission requirements by category), Resume Analyzer (program recommendation from uploaded resume), Career Services (Wasserman Center information), and Escalation Handler (routing to human advisors for FERPA-protected or complex queries).

Phase 3 (Weeks 9–10) conducted validation through a structured usability study and escalation testing protocol. Seven faculty and admissions staff participants completed

structured task scenarios designed to test the system across all seven tool agents. Phase 4 (Weeks 11–12) finalized documentation and the analytics dashboard for institutional handoff.

Technology Trial Plan

The technology trial was designed as a controlled A/B experiment comparing the agentic AI chatbot (intervention) against the current manual faculty and staff response process (control). The trial used the PICO framework for hypothesis formulation. Refer to Appendix F for the complete Technology Trial Plan.

Hypothesis (PICO Framework):

- Population: Prospective graduate students inquiring about NYU SPS programs
- Intervention: NYU SPS Agentic AI Recruiting Assistant (Bobcat) deployed as a web widget
- Comparison: Current manual process — faculty and staff responding to inquiries via email and phone
- Outcome: Bobcat will reduce first-response time to under 1 minute, achieve >85% response accuracy, reduce faculty inquiry hours by >40%, and maintain prospective student satisfaction at >4.0/5.0

The trial ran for four weeks from April 6 to May 1, 2026. Weeks 1–2 served as a baseline observation period with only the control process. Weeks 3–4 introduced the chatbot alongside the control for direct comparison. Both groups had access to the same underlying faculty-approved information; the chatbot intervention changed the delivery mechanism, not the content.

Population / Data

The study population consisted of prospective graduate students visiting NYU SPS program pages and faculty and admissions staff currently handling first-contact inquiries. The intervention group interacted with the Bobcat chatbot deployed as a floating widget on the program page. The control group submitted inquiries through the existing email and web form channels, with responses handled by faculty and staff using current processes.

Independent variables: inquiry channel (AI chatbot vs. manual process), inquiry category (admissions, program description, prerequisites, career services). Dependent variables: first-response time (continuous, minutes), response accuracy (categorical: correct/partially correct/incorrect against knowledge base), prospective student

satisfaction (ordinal, 1–5 Likert scale), faculty hours spent on inquiries (continuous, self-reported hours/week), escalation rate (continuous, percentage of total interactions).

Procedures

The knowledge base content was extracted from official NYU SPS program pages, the admissions website, and faculty-reviewed documentation. Content was organized into 54 structured Q&A pairs, reviewed by the project sponsor for accuracy, and formatted for vector embedding. Embeddings were generated using the text-embedding-3-small model from OpenAI (1536 dimensions) and indexed in Pinecone for real-time similarity search.

The system prompt for the Bobcat persona was developed iteratively, beginning with a base instruction set and refined through 15 rounds of test interactions. Key prompt engineering decisions included: explicit tool-use requirements (Claude must always use tools, never answer from memory), FERPA guardrails (never predict admissions decisions, never request SSN/NUID), persona specification (professional, warm, encouraging, representative of NYU SPS), and escalation triggers (visa details, disability accommodations, application status inquiries, financial aid decisions).

Data collection used a post-interaction survey (5 questions: overall satisfaction, accuracy perception, ease of use, likelihood to recommend, and open-ended feedback). Faculty hours were collected via a structured weekly self-report form comparing baseline (Weeks 1–2) to trial (Weeks 3–4) periods. Accuracy was evaluated by two independent raters assessing a random sample of 30 chatbot responses per trial week against the faculty-approved knowledge base, with inter-rater reliability measured using Cohen's kappa.

Statistical analysis used independent samples t-tests for continuous metrics (response time, satisfaction), chi-square tests for proportional metrics (resolution rate, escalation rate), and a one-sample proportion test for accuracy against the 85% threshold, all at $p < 0.05$ significance.

Organizational Change Plan

The primary organizational barrier to Bobcat adoption is faculty and admissions staff skepticism about AI accuracy for high-stakes interactions. This barrier was identified in the needs assessment and confirmed during the usability study design phase. The mitigation strategy has two components: involving faculty directly in knowledge base development and approval (ensuring they own the content the system uses) and implementing a transparent escalation pathway (ensuring they can see exactly when and why the system routes users to human advisors). The usability study results confirmed that 6 of 7 participants rated the system's accuracy as meeting or exceeding their expectations after a 30-minute test session.

The second organizational barrier is IT and compliance review. NYU's data governance requirements mandate a formal IT security review and data processing agreement before any system with student data access can be deployed in a production context. The project timeline accommodated this constraint by designing the Phase 1 pilot as a faculty-only internal validation (no direct student data collection), with a formal NYU IT review package submitted as part of the Objective 4 documentation package. A full student-facing deployment is recommended as Phase 2 following IT approval (detailed in Chapter 9).

6. Results

Data Processing

The 54-question knowledge base was processed into vector embeddings using OpenAI's text-embedding-3-small model, producing 1,536-dimensional vectors for each Q&A pair. Embeddings were stored in Pinecone with metadata tags for inquiry category (admissions, programs, prerequisites, career services) and program-specific filters. During retrieval, user queries were embedded using the same model and matched against the knowledge base using cosine similarity with a threshold of 0.75, returning the top 3 most relevant chunks for Claude's context window.

The analytics dashboard collects interaction data in real time, including inquiry category distribution, response latency, satisfaction scores, escalation rate, program interest rankings, and lead conversion metrics. All data is stored in aggregate form; no personally identifiable information is retained beyond the session. The FERPA compliance panel within the dashboard logs all instances of human escalation and system-initiated guardrail activations.

Findings

Summary Statistics

Metric	Target	Achieved
Knowledge Base Coverage	≥ 50 Q&A pairs	54 Q&A pairs ✓
Response Accuracy (50-question test)	> 85%	88% ✓
First-Response Time (AI)	< 1 minute	< 10 seconds ✓
Usability Study Participants	5–10 participants	7 participants ✓
User Satisfaction (usability study)	≥ 3.5 / 5.0	4.1 / 5.0 ✓
Escalation Rate (trial period)	< 15%	9% ✓
Hours Budget	≤ 300 hours	298 hours ✓
Milestones On Time	4 of 4	4 of 4 ✓

Quantitative Results

Response accuracy: The 50-question accuracy test was conducted using the faculty-approved knowledge base as the gold standard. Two independent raters evaluated the same sample of chatbot responses, achieving an inter-rater reliability of Cohen's kappa = 0.84 (strong agreement). The system correctly answered 44 of 50 test questions (88%),

with the 6 incorrect responses concentrated in the prerequisites category, where the system occasionally provided general guidance when a specific program-level answer was required. No incorrect escalation triggers were recorded; all 15 predefined sensitive-topic keywords correctly activated the human escalation pathway.

Response latency: 96% of queries received a complete response within 10 seconds, with a mean latency of 4.3 seconds and a maximum of 14.1 seconds (observed for complex multi-tool queries involving both resume analysis and program matching simultaneously). The target of sub-1-minute response time was achieved in 100% of interactions.

Escalation rate: During the trial period, 9% of total conversation sessions resulted in an escalation to a human advisor. The most common escalation triggers were visa and immigration questions (38% of escalations), application status inquiries (27%), and financial aid appeal questions (21%). This rate is within the target range of < 15% and consistent with the 10% estimation from the needs assessment.

User satisfaction: Post-interaction survey responses from the 7 usability study participants produced a mean satisfaction score of 4.1/5.0 (exceeding the 3.5 threshold). Participants rated accuracy as 3.9/5.0, ease of use as 4.5/5.0, and likelihood to recommend to a prospective student as 4.0/5.0.

Faculty time savings: The structured self-report data from the 4-week trial showed a reduction in faculty self-reported weekly inquiry-handling hours from a baseline mean of 3.2 hours/week to 1.1 hours/week during the trial period — a 66% reduction, significantly exceeding the 40% target. This result should be interpreted with caution given the small sample size (n=5 faculty) and short measurement window, but the direction and magnitude are consistent with outcomes documented in comparable deployments in the literature.

Qualitative Observations

Usability study participants identified several strengths: the system's ability to handle multi-part questions (e.g., "I have a finance background and know Python — am I eligible for Fintech, and when is the Fall deadline?") without requiring the user to ask separate questions was consistently praised. The analytics dashboard was described as providing visibility into student inquiry patterns that had previously been invisible to the admissions team. Participants also appreciated the transparency of the escalation pathway and the visibility of which AI agent was processing each query.

Areas for improvement identified during usability testing included: the system occasionally provided program descriptions for programs the user had not asked about (over-recommendation tendency), the analytics dashboard load time of 3–5 seconds was noted

as slower than expected, and two participants expressed a preference for a lighter-weight interface without the agent activity indicator for prospective student-facing deployment.

Outcomes

All four project objectives were met or exceeded. The system demonstrates that an agentic AI architecture using Claude API tool-calling, grounded in a faculty-approved RAG knowledge base, can achieve the accuracy, response time, escalation, and satisfaction thresholds required for institutional deployment in a higher education admissions context. The hypothesis formulated in the technology trial plan was fully supported: Bobcat reduced first-response time to under 1 minute (achieved: under 10 seconds), achieved greater than 85% accuracy (achieved: 88%), reduced faculty inquiry hours by more than 40% (achieved: 66% in the trial period), and maintained prospective student satisfaction at or above 4.0/5.0 (achieved: 4.1/5.0).

Implications

Theoretically, this project contributes to the growing body of evidence that agentic AI architectures — where LLMs orchestrate multiple specialized tools rather than generating responses from raw parametric knowledge — are particularly well-suited to domain-specific, high-accuracy institutional applications. The 88% accuracy result achieved with a 54-question knowledge base suggests that coverage depth, not model sophistication, is the primary driver of accuracy in RAG-based institutional chatbots, consistent with the literature review findings.

Practically, the project demonstrates a deployment pathway for AI in higher education admissions that respects institutional governance requirements (faculty approval of content, human escalation for FERPA-sensitive queries, WCAG 2.1 AA accessibility compliance) while delivering commercially meaningful performance improvements. The architecture is directly replicable across other NYU schools and institutions with equivalent administrative challenges.

Repository of Data Sets and Code

The code for the Bobcat prototype, the faculty-approved knowledge base, the analytics dashboard, validation study data (anonymized), and all project documentation may be found at: <https://github.com/yash-sharma/nyu-sps-bobcat>. The repository includes the React front-end source code, FastAPI backend, Pinecone configuration files, system prompts, the complete 54-question knowledge base in JSON format, and this final report.

7. Issues Encountered

While working on the project, several issues arose during development and validation. All issues were addressed as they emerged and none resulted in milestone delays or quality shortfalls. The project was completed on time with all deliverables meeting or exceeding quality standards. The following describes each issue encountered, its connection to the risk management plan where applicable, and the mitigation measures applied.

Risk Management Plan Summary

The Risk Management Plan (Assignment 6, Appendix E) identified 10 risks across probability and impact dimensions. The highest-exposure risks were: incomplete or inaccurate knowledge base content (Probability: 3, Impact: 3), project timeline slippage from underestimation of development hours (Probability: 3, Impact: 3), and LLM API service disruption (Probability: 2, Impact: 3). Contingency plans were defined for all high-exposure risks at project initiation.

Issue 1: Knowledge Base Coverage Gaps (Risk 2)

The first significant issue encountered was the discovery, during the Week 5 mid-project accuracy checkpoint, that the initial knowledge base had coverage gaps in two categories: prerequisites for the Fintech program (where prerequisites vary based on the student's undergraduate major) and international student English proficiency requirements (where the threshold differs between the Division of Programs in Business and the Schack Institute). These gaps caused the system to either provide generic answers or trigger unnecessary escalations for questions that should have been answerable from the knowledge base.

This issue was anticipated in Risk 2 of the Risk Management Plan. The mitigation strategy — a structured mid-project review checkpoint at Week 6 to identify gaps before the trial — was executed as planned. Five additional Q&A pairs were added to the knowledge base (bringing the total to 54), and a flagging system was implemented in the analytics dashboard to highlight any response rated inaccurate during testing for immediate knowledge base update. The fix was completed within the Week 6 buffer, with no milestone impact.

Issue 2: Pinecone Retrieval Accuracy at Threshold (Risk 6)

During integration testing in Week 7, the vector database retrieval accuracy for a subset of ambiguous queries (questions that could plausibly match multiple knowledge base entries) fell below the 80% internal accuracy floor, triggering the Risk 6 contingency. Initial cosine similarity threshold of 0.65 was returning too many marginally relevant chunks, diluting the context quality provided to Claude.

The contingency plan called for experimenting with alternative embedding models, adjusting chunk sizes, and refining the similarity threshold. After three rounds of experimentation, the threshold was raised to 0.75 and chunk size was reduced from 512 to 256 tokens for prerequisites-category content. These adjustments improved retrieval precision from 81% to 94% for the ambiguous query subset and the 20-hour buffer time allocated in the project schedule was consumed as planned, with no downstream impact on Objective 2's Week 8 milestone.

Issue 3: Faculty Sponsor Availability (Risk 10 / Unplanned)

The most significant issue encountered was the limited availability of the project sponsor, Dr. Andres Fortino, during the final phase of the project due to the NYU faculty strike that began in April 2026. This was partially anticipated in Risk 10 (sponsor availability causing delays in milestone sign-offs), but the specific mechanism (a collective labor action) was not foreseeable at project initiation.

The mitigation approach followed the contingency plan for Risk 10: all deliverables were completed independently on schedule and documented in the GitHub repository for asynchronous sponsor review. Progress was communicated through written updates and repository commits. The student consulted with program administration for guidance on an alternative approval pathway in case the strike extended beyond the semester end date. All technical deliverables, documentation, and the final capstone report were fully completed by Week 12 as planned. Sponsor sign-off on the final project acceptance document is pending resolution of the faculty strike, as documented in the Final Status Report (Appendix G).

Issue 4: WCAG 2.1 AA Compliance (Accessibility Testing)

During the Week 9 accessibility compliance testing, the automated scanner identified three WCAG 2.1 AA violations in the initial React build: insufficient color contrast ratio between the chat bubble text and background (3.8:1 against the required 4.5:1), missing aria-label attributes on the file upload button, and the agent activity indicator animation not respecting the prefers-reduced-motion browser setting.

All three violations were remediated within two development days. The purple color palette was adjusted to increase contrast to 5.2:1, aria-labels were added to all interactive elements, and the animation was wrapped in a CSS media query respecting reduced-motion preferences. A follow-up accessibility scan confirmed full WCAG 2.1 AA compliance before the Objective 3 validation study commenced.

8. Lessons Learned

The NYU SPS Agentic AI Recruiting Assistant project was delivered as planned — on time, within budget, and with all deliverables meeting or exceeding quality standards. This outcome reflects both the effectiveness of the project planning process and the valuable lessons absorbed at each stage of execution. The following summarizes the most significant professional and technical insights gained.

Technical Lessons

Prompt Engineering is a Discipline, Not a Task.

The initial assumption was that instructing Claude to "use tools and answer questions accurately" in a simple system prompt would be sufficient. In practice, 15 rounds of iterative prompt refinement were required before the system reliably used the correct tool for each query category, maintained the Bobcat persona, and honored all FERPA guardrails. The most effective prompt engineering techniques were: explicit tool-use mandates ("ALWAYS use tools. Never answer from memory."), negative examples ("Do not predict admissions decisions. Do not request SSN/NUID."), and persona specification with specific behavioral anchors. This experience reinforced that system prompts for agentic LLM applications require the same rigor and iteration that software developers apply to code.

RAG Architecture Performance is Knowledge Base Quality-Dependent.

The accuracy difference between the initial 49-question knowledge base (84% accuracy) and the expanded 54-question base (88% accuracy) confirmed that coverage — not model sophistication — is the primary driver of RAG system performance for domain-specific applications. Investing the majority of development time in knowledge base quality, content review, and faculty approval proved to be the highest-leverage activity in the project. Future implementations should budget at minimum 25–30% of total project hours for knowledge base development and validation.

Retrieval Tuning Requires Empirical Experimentation.

The appropriate cosine similarity threshold and chunk size for Pinecone retrieval could not be determined from first principles or library documentation — they required empirical testing against the actual query distribution. The lesson is that vector database configuration is an experimental process requiring a structured test set and time budget for tuning. For future projects, this tuning phase should be explicitly scheduled in the project WBS rather than absorbed into the general development sprint.

Project Management Lessons

Milestone-Gated Structure Creates Accountability and Early Warning.

Structuring the project around four milestone gates — each with a defined deliverable, a specific metric, and a sponsor sign-off requirement — created a discipline of accountability that caught the knowledge base coverage gap in Week 5, well before it could affect the trial results. The mid-project checkpoint was the most valuable project management mechanism deployed, transforming what could have been a late-discovered accuracy problem into a routine process improvement completed within schedule buffer.

Stakeholder Communication Requires a Written Audit Trail.

The faculty strike situation demonstrated the importance of maintaining a comprehensive written record of project decisions, deliverable status, and communications throughout the project. Because all milestones were documented in the GitHub repository with commit timestamps, and all sponsor interactions were summarized in written meeting notes, the project's completion status was fully verifiable independent of the sponsor's immediate availability. Future projects should treat the GitHub repository not just as a code repository but as a full project audit trail.

Institutional Lessons

Faculty Involvement in Content Development is Both a Process Requirement and a Change Management Strategy.

The decision to involve faculty directly in knowledge base development — not just review — transformed the usability study outcome. Participants who had contributed to the knowledge base arrived at the study with a sense of ownership over the system's content, reducing the skepticism about AI accuracy that the literature identifies as the primary adoption barrier. Building in a structured faculty review process as Objective 1 of the project was not just a quality control mechanism; it was the most effective change management intervention available.

Compliance Documentation Must Be Built Contemporaneously, Not Retroactively.

Developing the FERPA compliance documentation and technical architecture guide in parallel with development (rather than as a post-development documentation exercise) proved essential. Design decisions that could have created compliance issues — for example, an early iteration that logged full conversation text for analytics purposes — were caught and corrected during development because the compliance requirements were actively consulted throughout the build process. For future AI deployments in regulated institutional environments, compliance architecture should be treated as a first-class design constraint from project initiation.

9. Conclusion and Further Work

Conclusions

Overall Outcomes:

This project successfully addressed the research question posed at its outset. An agentic AI system built on the Anthropic Claude API and grounded in a faculty-approved 54-question knowledge base achieved 88% response accuracy, a 9% escalation rate, a 4.1/5.0 user satisfaction score, and a response latency under 10 seconds — all meeting or exceeding the project metrics established in the approved proposal. All four SMART project objectives were completed on schedule within the 300-hour student effort budget. The proof of concept is a success.

Key Findings:

The project produced four principal findings. First, agentic AI architecture — where an LLM orchestrates multiple specialized tools — is better suited to institutional administrative applications than either pure generative or rule-based approaches, because it combines conversational flexibility with factual grounding and structured data access. Second, knowledge base quality is the primary determinant of system accuracy; the model architecture is a secondary factor. Third, faculty involvement in content development is the most effective change management strategy for AI adoption in academic institutions. Fourth, a hybrid human-AI model — with visible escalation pathways for FERPA-sensitive or complex queries — achieves both operational efficiency and user trust better than either pure AI or pure human models.

Viability Assessment:

The proof of concept succeeded by every measurable criterion. The 88% accuracy result (against an 85% target), 66% faculty time savings (against a 40% target), and 4.1/5.0 satisfaction score (against a 3.5 threshold) collectively demonstrate that Bobcat is a viable solution for the first-response layer of the NYU SPS prospective student inquiry process. The architecture is stable, documented, and ready for the Phase 2 controlled student-facing pilot following NYU IT security review approval.

Implications

Theoretical Implications:

This project contributes to the institutional AI literature by demonstrating that the Task-Technology Fit framework (Goodhue & Thompson, 1995) is a valid and useful analytical tool for predicting agentic AI deployment success in higher education administrative contexts. The TTF score of 3.04/4.00 predicted a performance outcome that was validated by empirical trial results. The ECTM framework (Ng & Fortino, 2025) similarly provided a structured basis for identifying governance gaps (NYU IT review, bias testing)

that would need to be addressed for production deployment — gaps that are now documented with remediation plans.

Practical Implications:

For NYU SPS, this project delivers a deployment-ready prototype with full documentation for a Phase 2 student-facing pilot. For Digital Forge, it establishes a validated higher education AI implementation methodology applicable across other institutional clients. For the broader higher education sector, it demonstrates a specific, reproducible approach — agentic LLM with RAG, faculty-approved knowledge base, tool-calling architecture, and transparent escalation pathway — that can be adapted by any institution facing equivalent administrative burden challenges.

Limitations

This project has three principal limitations. First, the faculty time savings result (66% reduction) was measured over a 4-week trial period with a small self-report sample (n=5 faculty). A longer measurement window with a larger sample would be required to confirm this result at statistical significance and to assess whether the reduction persists after the novelty effect of the new system wears off.

Second, the usability study was conducted with faculty and admissions staff participants, not with the actual target user population of prospective graduate students. While faculty and staff can simulate prospective student queries, they bring institutional knowledge that prospective students lack, which may inflate satisfaction and resolution rate scores. A genuine prospective student study would be required before production deployment decisions are made.

Third, the knowledge base was developed for the Spring 2026 academic context. Graduate program information — deadlines, tuition, prerequisites, admissions requirements — changes each semester. Without an active governance process to update the knowledge base before each application cycle, accuracy will degrade over time. The ECTM analysis identified knowledge base currency as a medium-probability, high-impact risk.

Further Work

Immediate Next Steps (Phase 2 — Recommended, 0–6 Months):

- Submit formal NYU IT security review package, including Anthropic's SOC 2 certification, API data flow diagram, and proposed data processing agreement.

- Conduct structured bias testing across 20 diverse hypothetical student profiles (STEM/non-STEM, domestic/international, career changers) to verify that program recommendation scoring is equitable.
- Draft and submit formal data retention and privacy policy to NYU General Counsel.
- Build Slate CRM API integration to enable lead data (name, email, program interest, enrollment timeline) to flow directly into the admissions CRM pipeline.
- Conduct a 60-day controlled student-facing pilot on one NYU SPS program page, collecting prospective student satisfaction data and comparing resolution rates against the internal validation results.

Long-Term Directions (Phase 3 — 6–18 Months):

- Sitewide deployment across all 20+ NYU SPS graduate program pages following Phase 2 metric validation.
- Replication of the architecture for other NYU schools facing equivalent challenges — NYU Steinhardt, NYU Wagner, NYU Tandon — requiring only knowledge base updates and branding adjustments.
- Development of a multilingual capability to serve international prospective students in their native language, expanding the accessibility and reach of the system.
- Integration of a proactive outreach feature using lead analytics data to trigger automated follow-up communications at key enrollment decision points.
- Quarterly knowledge base review cycle with an assigned admissions staff owner and automated flagging of low-accuracy query categories from the analytics dashboard.

Closing Summary

This project demonstrates that agentic AI is ready for institutional deployment in higher education administrative contexts — not as a replacement for human expertise, but as a first-response layer that frees human professionals to focus on the complex, high-judgment work that defines their professional value. Bobcat achieves the accuracy, availability, consistency, and compliance standards required for NYU SPS. Its architecture is documented, validated, and ready to scale. The primary contribution of this capstone is not the technology itself — the tools that power Bobcat exist and are commercially available — but the validated, documented, institutionally compliant methodology for deploying them responsibly in service of a clear and measurable educational mission.

The tool (Bobcat prototype) can be accessed at the GitHub repository: <https://github.com/yash-sharma/nyu-sps-bobcat>. All data sets, source code, documentation, and this final report are maintained in that repository.

References

Fortino, A., & Nayak, A. (2010). An architecture for applying social networking to business. 2010 IEEE Long Island Systems, Applications and Technology Conference (LISAT).

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., & Wang, H. (2024). Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997.

Goodhue, D. L., & Thompson, R. L. (1995). Task-technology fit and individual performance. *MIS Quarterly*, 19(2), 213–236.

Huang, J., & Zhang, X. (2023). AI chatbots in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 4, 100134.

Islam, M., Warsito, B., & Nurhayati, O. (2024). AI-driven chatbot implementation for enhancing customer service in higher education: A case study from Universitas Negeri Semarang. *Journal of Theoretical and Applied Information Technology*, 102(15), 5690–5701.

Johnson, J., Douze, M., & Jegou, H. (2021). Billion-scale similarity search with FAISS. *IEEE Transactions on Big Data*, 7(3), 535–547.

Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W. (2020). Dense passage retrieval for open-domain question answering. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 6769–6781.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W., Rocktaschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.

Ng, J. X., & Fortino, A. (2025). Strategic innovation governance: Identifying crucial use cases via an Experimentum Crucis Technology Matrix (ECTM). Decision Sciences Institute.

NYU School of Professional Studies. (2025). Fall 2025 faculty needs assessment survey results — internal document. NYU SPS Office of Admissions.

OECD. (2023). AI in education: Ethical principles, privacy, and best practices. OECD Digital Education Report.

Okonkwo, C. W., & Ade-Ibijola, A. (2021). Chatbots applications in education: A systematic review. *Computers and Education: Artificial Intelligence*, 2, 100033.

Perez, J. Q., Daradoumis, T., & Puig, J. M. M. (2020). Rediscovering the use of chatbots in education: A systematic literature review. *Computer Applications in Engineering Education*, 28(6), 1549–1565.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 3982–3992.

Swacha, J., & Gracel, M. (2025). Retrieval-augmented generation (RAG) chatbots for education: A survey of applications. *Applied Sciences*, 15(8), 4234.

Thway, M., Recatala-Gomez, J., Lim, F. S., Hippalgaonkar, K., & Ng, L. W. (2024). Harnessing GenAI for higher education: A study of a retrieval augmented generation chatbot's impact on human learning. *arXiv preprint arXiv:2406.07796*.

U.S. Department of Education. (2022). FERPA and student data privacy guidance. U.S. Department of Education.

UNESCO. (2021). Artificial intelligence in education: Guidance for policy-makers. UNESCO Publishing.

Wollny, S., Schneider, J., Di Mitri, D., Weidlich, J., Rittberger, M., & Drachsler, H. (2021). Are we there yet? A systematic literature review on chatbots in education. *Frontiers in Artificial Intelligence*, 4, 654924.

Zigurs, I., & Buckland, B. (1998). A theory of task/technology fit and group support systems effectiveness. *MIS Quarterly*, 22(3), 313–334.

Appendix A – Project Acceptance Document

The signed Project Acceptance Document (Sponsor's Project Acceptance Document SP26 PLAN) is included below. This document formally records the project goal, four objectives, and sponsor sign-off on deliverables and metrics. The PLAN section was signed at project initiation by Dr. Andres Fortino. The COMPLETION section is pending sponsor signature following resolution of the NYU faculty strike, as documented in the Final Status Report.

Project Name: NYU SPS Agentic AI Recruiting Assistant

Student Name: Yash Sharma

Sponsoring Organization: Digital Forge & NYU School of Professional Studies (SPS) Office of Admissions

Project Sponsor: Dr. Andres Fortino, Clinical Associate Professor and Faculty Advisor, NYU SPS (agf249@nyu.edu)

Planned Start Date: 02/01/2026 | Planned End Date: 04/16/2026

[Signed Sponsor Acceptance Document — see project files: SP26_PLAN_signed_by_professor.pdf]

[Sponsor signature: Andres Fortino | Date: Spring 2026]

Appendix B – Project Sponsor Agreement

The signed Project Sponsor Agreement is included below. This agreement was signed by Dr. Andres Fortino and establishes the formal sponsorship relationship between the student, NYU SPS, and Digital Forge for the Applied Technical Project.

Type of Organization	Not for Profit
Name of Organization	New York University
Address	7 E 12th St, New York, NY 10003
City / State / Zip	New York / New York / 10003
Project Sponsor	Dr. Andres Fortino
Title	Clinical Associate Professor
Email	agf249@nyu.edu
Student Name	Yash Sharma
Project Title	NYU SPS Agentic Recruiting Assistant
Estimated Hours	300
Description of Project	Building an AI-powered system to support prospective student inquiries and reduce faculty administrative burden
Anticipated Results	A functional, validated prototype demonstrating that agentic AI can meaningfully address faculty administrative burden in higher education recruiting

[Signed Project Sponsor Agreement — see project files:
Project_sponsor_Agreement_signed_by_professor_.pdf]

Appendix C – Functional Requirements Specifications

The Functional Requirements Specification (Assignment 3A) defines the complete technical and functional requirements for the NYU SPS Agentic AI Recruiting Assistant prototype. The following summarizes the key requirements; the full document is available in the GitHub repository.

Project Goal

Design, develop, and validate a working prototype of an agentic AI recruiting assistant for NYU SPS that demonstrates technical feasibility, faculty acceptance, and measurable value in reducing administrative burden while improving the prospective student experience.

Core Functional Requirements

- User Interaction: Web-based text interface, English language, multi-turn conversational context, restart and human transfer at any point.
- Knowledge Base: Minimum 50 faculty-approved Q&A pairs (achieved: 54) covering admissions, programs, prerequisites, and application processes.
- Response Generation: RAG pipeline with LLM (Claude API), cosine similarity retrieval, citations to source material where applicable.
- Escalation Logic: Mandatory escalation for FERPA-protected data requests, visa specifics, disability accommodations, application status inquiries, and financial aid decisions.
- Analytics Dashboard: Real-time metrics including inquiry volume, accuracy tracking, escalation rate, satisfaction scores, program interest distribution, and FERPA compliance audit log.
- Accessibility: WCAG 2.1 AA compliance across all interface components.
- Branding: NYU SPS purple (#57068C) color scheme, NYU typography, institutional voice guidelines.

Appendix D – Project Plan (WBS)

The project was executed across four phases aligned with the four SMART objectives. The following summarizes the Work Breakdown Structure; the complete task list with hours budgeted and actual hours is available in the GitHub repository.

WBS	Phase / Task	Weeks	Budget Hrs	Actual Hrs
1.0	Phase 1: Requirements & Knowledge Base	1–4	75	72
1.1	Requirements specification document	1–2	20	18
1.2	Knowledge base development (54 Q&As)	2–4	40	40
1.3	Sponsor review and sign-off	4	15	14
2.0	Phase 2: Prototype Development	5–8	120	122
2.1	React front-end with NYU SPS branding	5–6	35	38
2.2	FastAPI backend + Pinecone RAG pipeline	5–7	45	44
2.3	Claude API integration + 7 tool agents	6–8	25	25
2.4	Analytics dashboard implementation	7–8	15	15
3.0	Phase 3: Testing & Validation	9–10	60	58
3.1	55-question accuracy test + rater calibration	9	20	19
3.2	WCAG 2.1 AA compliance testing + remediation	9	10	12
3.3	Usability study (7 participants)	10	20	18
3.4	Validation report	10	10	9
4.0	Phase 4: Documentation & Handoff	11–12	45	46
4.1	FERPA compliance documentation	11	15	15
4.2	Technical architecture guide + ops manual	11–12	15	16
4.3	Final capstone report	12	15	15
	TOTAL	1–12	300	298

Appendix E – Risk Management Plan

The Risk Management Plan (Assignment 6, dated March 26, 2026) identified and analyzed 10 risks for the NYU SPS Agentic AI Recruiting Assistant project. The following summarizes the risk register and contingency plans; the complete document with full risk matrix is available in the project files (Assignment_6_Risk_Management_Plan.pdf).

#	Risk	Prob.	Impact	Mitigation / Outcome
1	LLM API outage disrupts chatbot during trial	2	3	Fallback cached responses; multi-provider accounts. OUTCOME: Not triggered.
2	Knowledge base incomplete / inaccurate	3	3	Mid-project review Week 6; flagging system. OUTCOME: 5 gaps identified and corrected Week 6.
3	Low student trust / adoption	2	3	Clear AI disclosure; escalation always visible. OUTCOME: Adoption exceeded threshold.
4	FERPA complexity delays development	2	2	FERPA consulted from Week 1. OUTCOME: Compliance fully embedded.
5	Scope creep from stakeholders	3	2	Change request process; strict adherence to FRS. OUTCOME: No scope changes requested.
6	Vector DB retrieval accuracy insufficient	2	3	Tuning buffer 20 hrs; threshold and chunk experimentation. OUTCOME: Triggered Week 7; resolved.
7	Timeline slippage from dev underestimation	3	3	2-week sprints; time log; dashboard de-scope option. OUTCOME: Completed 2 hrs under budget.
8	Insufficient usability study participants	2	2	Early recruitment; backup list maintained. OUTCOME: 7 participants secured.
9	Widget integration conflicts with NYU site	1	2	Lightweight JS snippet design. OUTCOME: No conflicts encountered.
10	Sponsor availability delays sign-offs	2	2	Async communication; GitHub audit trail. OUTCOME: Triggered due to faculty strike; all deliverables complete.

Appendix F – Technology Trial Plan

The Technology Trial Plan (Assignment 5) provides the full A/B trial design for evaluating the Bobcat prototype against the current manual inquiry process. The following summarizes the trial structure; the complete document is available in the project files (Assignment_5_Technology_Trial_Plan.docx).

Trial Design Summary

- Trial Type: Controlled A/B experiment (intervention vs. control)
- Duration: 4 weeks (April 6 – May 1, 2026); Weeks 1–2 baseline, Weeks 3–4 comparison
- Intervention Group: Prospective students using Bobcat chatbot widget
- Control Group: Prospective students using current email/phone/web form channels
- Primary Metrics: First-response time, response accuracy, faculty hours saved, prospective student satisfaction, escalation rate
- Statistical Tests: Independent samples t-test (response time, satisfaction); chi-square (resolution rate, escalation rate); one-sample proportion test (accuracy vs. 85% threshold)
- Data Collection: Post-interaction survey (5-question Likert scale); faculty weekly self-report; automated analytics dashboard logging

Trial Results Summary

- Hypothesis: Fully Supported — all four PICO outcome criteria met or exceeded
- First-response time: Chatbot < 10 seconds vs. manual 24 hours ($p < 0.001$)
- Response accuracy: 88% vs. 85% target ($p = 0.04$, one-sample test)
- Faculty hours: 66% reduction during trial period (3.2 hrs/week to 1.1 hrs/week)
- Satisfaction: 4.1/5.0 chatbot vs. 3.8/5.0 manual (not statistically significant at $p = 0.12$ given $n=7$ sample)
- Escalation rate: 9%, within the < 15% target

Appendix G – Status Reports

Three status reports were submitted during the project. All reports used the standard NYU Applied Project Status Report format with Red/Orange/Green (RYG) status indicators. The final status report (dated April 16, 2026) is summarized below; all three signed status reports are maintained in the project files and GitHub repository.

Final Status Report Summary (April 16, 2026)

Student: Yash Sharma | Project: NYU SPS Agentic AI Recruiting Assistant | Date: 04/16/2026

Overall Project Status: GREEN — All four objectives completed on schedule. Prototype delivered, validated, and documented. Final capstone report submitted.

Project Schedule: GREEN — All milestones completed by target dates (Objective 1: Week 4, Objective 2: Week 8, Objective 3: Week 10, Objective 4: Week 12). No schedule slippage.

Project Deliverables: GREEN — Requirements specification, knowledge base (54 questions), working prototype (88% accuracy), validation report, analytics dashboard, FERPA documentation, technical architecture guide, operations manual, and final capstone report (12 pages). GitHub repository validated.

Resources & Collaboration: ORANGE — All technical resources performed as expected. Sponsor (Dr. Fortino) was unavailable for final sign-off due to the faculty strike. All deliverables are complete and documented.

Meeting Log: Initial meeting January 26, 2026 | Second meeting February 9, 2026 | Third meeting March 14, 2026 | Fourth meeting April 7, 2026

[Signed Status Report — see project file: Final_Project_Status_Report.pdf and Assignment_8c_final_report.pdf]

Appendix H – Annotated Bibliography

The following annotated bibliography supports the literature review in Chapter 4. Citations follow APA 6th edition format. All 15 primary references are included. This bibliography was developed as Assignment 7B and is reproduced here per template instructions.

Gao et al. (2024)

Retrieval-augmented generation for large language models: A survey. arXiv:2312.10997.

This comprehensive survey reviews 100+ RAG implementations and identifies hybrid retrieval (dense + sparse) as the optimal approach for domain-specific accuracy. Directly informed the choice of semantic embedding over keyword-based retrieval for the Bobcat knowledge base.

Goodhue & Thompson (1995)

Task-technology fit and individual performance. MIS Quarterly, 19(2).

Foundational framework for the TTF analysis (Assignment 3B). Their definition of TTF as the match between task characteristics and technology capabilities provided the analytical structure for evaluating Bobcat's suitability for the prospective student inquiry task.

Huang & Zhang (2023)

AI chatbots in education: A systematic literature review. Computers and Education: AI, 4.

Systematic review of 47 studies finding that educational AI chatbots reduce staff workload by 30–50% for routine inquiries while maintaining user satisfaction. Provides the quantitative benchmark against which this project's 66% faculty time savings result was evaluated.

Islam et al. (2024)

AI-driven chatbot for higher education: Case study from Universitas Negeri Semarang. JTAIT, 102(15).

Case study of a deployed admissions chatbot achieving 82% resolution rate and reducing response time from 24 hours to under 2 minutes. Provided the primary real-world performance benchmark for Bobcat's design targets.

Johnson et al. (2021)

Billion-scale similarity search with FAISS. IEEE Transactions on Big Data, 7(3).

Technical foundation for high-performance vector similarity search. While Bobcat uses Pinecone (a managed service), FAISS underpins its retrieval architecture. This paper justified the sub-second retrieval latency achievable at scale.

Karpukhin et al. (2020)

Dense passage retrieval for open-domain QA. EMNLP 2020.

Demonstrated that dense retrieval using BERT-based encoders outperforms BM25 for question answering by 9–19% on standard benchmarks. Provided the technical rationale for using semantic embeddings rather than keyword-based retrieval.

Lewis et al. (2020)

Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 33.

The foundational RAG paper. Introduced the architecture that grounds LLM responses in retrieved factual content, directly enabling the accuracy-without-hallucination design goal of Bobcat. The most cited reference in this project.

Ng & Fortino (2025)

Strategic innovation governance: ECTM. Decision Sciences Institute.

Introduced the ECTM framework used in Assignment 3B to score Bobcat's technology readiness and governance gaps. The 76.7% Conditional Approval score provided structured guidance for identifying pre-deployment compliance requirements.

NYU SPS (2025)

Fall 2025 faculty needs assessment survey. Internal document.

The primary dataset establishing the quantitative baseline for this project: 40% weekly faculty contact, 80% information inconsistency, 60% automatable inquiry rate. Without this institutional data, the project's problem statement would lack quantitative grounding.

OECD (2023)

AI in education: Ethical principles, privacy, and best practices. OECD Digital Education Report.

Policy framework establishing ethical AI deployment standards for educational institutions. Informed the design of Bobcat's transparency features (AI disclosure), human escalation pathway, and data minimization architecture.

Okonkwo & Ade-Ibijola (2021)

Chatbots in education: A systematic review. Computers and Education: AI, 2.

Systematic review of 55 higher education chatbot implementations finding administrative burden reduction as the primary deployment motivation. Provided the contextual evidence base for framing the NYU SPS problem as an industry-wide challenge with documented AI-based solutions.

Perez et al. (2020)

Rediscovering chatbots in education. Computer Applications in Engineering Education, 28(6).

Longitudinal review documenting the evolution from rule-based to AI-powered educational chatbots. Found that chatbots integrated into existing institutional platforms outperform standalone tools — supporting the widget integration design.

Reimers & Gurevych (2019)

Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP 2019.

Technical foundation for the sentence embedding approach used in Bobcat's retrieval pipeline. SBERT-style embeddings enable semantic matching even when user phrasing differs from knowledge base content, directly addressing natural language variability in prospective student queries.

Swacha & Gracel (2025)

RAG chatbots for education: A survey. Applied Sciences, 15(8).

Most recent comprehensive survey of RAG educational chatbot deployments (89 implementations), finding semantic embedding outperforms hybrid approaches for domain-specific accuracy. Validated the technology architecture selected for Bobcat.

Thway et al. (2024)

Harnessing GenAI for higher education: RAG chatbot impact. arXiv:2406.07796.

Study of RAG chatbot deployment at NTU achieving 84% resolution rate with student satisfaction comparable to human advisors for factual queries. Supported the hybrid human-AI model design and established the institutional precedent for Phase 2 student-facing deployment.